

TECHNICAL REPORT

Person-Centric Representations for Four- and Nine-Class POLAR Posture Recognition

Source-audited data, complementary classifiers and locked evaluation

Author	Abdulla Huseyinli
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Repository
<https://github.com/abdullahuseyinli-dot/polar-posture-recognition>

Person-Centric Representations for Four- and Nine-Class POLAR Posture Recognition

Abstract

This study evaluates person-centric adaptation and complementary visual representations for still-image posture classification. A source-overlap audit retains 35,007 of the original 35,324 POLAR images without reassigning official split membership. Frozen DINOv2, DINOv3, SigLIP2 and ConvNeXt V2 classifiers are compared with partially adapted models and development-fixed probability fusions. The development-nominated conservative fusion obtains 95.211% macro-F1 on 3,329 four-class test images and 94.583% on 6,984 nine-class images. Paired gains over the historical four-class ensemble and the nine-class adapted-DINOv2 reference are +1.223 and +0.679 percentage points, respectively. Smaller gains against the immediate prior models fail the complete locked promotion rule, leaving those priors retained. Public probability arrays support replay of candidate metrics, paired comparisons and resampling without images or model weights. The study is an audited-cohort benchmark, not an external state-of-the-art claim; the previously inspected four-class test forms part of the nine-class test.

1. Scope and contribution

The task is posture classification given an RGB image and a supplied target-person bounding box. It is not person detection, temporal action recognition or clinical assessment. Four classes cover sitting, standing, walking and running; the nine-class task adds bending, jumping, lying, squatting and stretching.

The contribution is an empirical system and evaluation protocol: source-aware data preparation, controlled frozen-feature comparisons, person-preserving adaptation, fixed complementary fusion and traceable prediction-level evidence. The pretrained encoders are prior work, not newly invented backbones. This report extends the [original four-class study](#); its historical five-component ensemble remains a fixed comparator.

2. Dataset and evaluation boundary

POLAR version 1 provides the original data and labels. See the [dataset record](#) and [Ma and Liang's original paper](#). The audit inherited the original four-class quarantine and added label-blind canonical-source and confirmed perceptual-hash checks across all nine classes.

Population	Train	Validation	Test	Total
Original nine-class release	21,194	7,065	7,065	35,324
Source-audited nine-class cohort	21,057	6,966	6,984	35,007
Source-audited four-class subset	9,958	3,327	3,329	16,614

In total, 317 images in 154 cross-split source components were quarantined, including 125 inherited exclusions. No image was moved to a different official split. Final models were refitted on train plus validation after selection was locked. The public [cohort package](#) records membership, labels, boxes, source groups and image/annotation hashes.

The audit is not a verified subject, scene or session split. Same-split near duplicates are not exhaustively grouped, and the nine-class audit did not include a new exhaustive embedding-based duplicate search. Pretraining overlap is unknown. The historically inspected four-class test is nested in the nine-class test; the final phase is locked but its test population is not wholly untouched.

3. Methods

3.1 Frozen two-view classifiers

Each frozen encoder provides full-frame and 10%-context person-view embeddings, which are concatenated. A training-only scaler and unweighted RBF classifier use $C = 10$ and $\gamma = 1/d$. Sigmoid calibration uses five source-grouped training folds. The four families share the downstream search budget and split membership, not matched preprocessing, feature width, pretraining data, model capacity or total computation. DINOv2 combines four normalized CLS tokens and a final-layer mean patch token (7,680 dimensions across two views). DINOv3 and SigLIP2 use pooled features (1,536 combined dimensions); ConvNeXt V2 uses 2,048. DINOv2 and ConvNeXt V2 resize the short edge to 256 then center-crop to 224 with bicubic interpolation; DINOv3 and SigLIP2 resize directly to 224×224 with bilinear interpolation. SigLIP2 uses per-channel mean and standard deviation 0.5; the others use ImageNet normalization. The [feature-contract table](#) records these distinctions.

The DINOv3-B checkpoint was recovered from the transferred project and verified by hash. Its original processor JSON was unavailable; Transformers 5.5.3 defaults were reconstructed at 224 pixels with bilinear resizing and ImageNet normalization. This establishes the local recipe, not provider preprocessing byte parity or the general 256-pixel DINOv3 recipe. DINOv3-L was unavailable and is not reported.

3.2 Person-preserving adaptation

Adapted models use a 25%-context person view, padded to square and then resized to 224×224 , preserving aspect ratio. DINOv2 adapts its last four blocks and backbone normalization layers; SigLIP2 its last four blocks, final normalization and attention pooler; ConvNeXt V2 its last stage and final normalization. Training uses unweighted cross-entropy, AdamW, head/backbone learning rates 0.001/0.000005, weight decay 0.0001, dropout 0.1, gradient clipping 1 and effective batch size 64 (16 with accumulation 4). Neural fits use CUDA bfloat16 and seeds 42, 52 and 62. No mixup or label smoothing is used in the locked recipe.

Refit epoch counts are medians selected in development: four-class SigLIP2 10 and ConvNeXt V2 15; nine-class DINOv2 16, SigLIP2 10 and ConvNeXt V2 18. The learning-rate schedule retains its original 20-epoch horizon. Final fits do not select epochs using test performance.

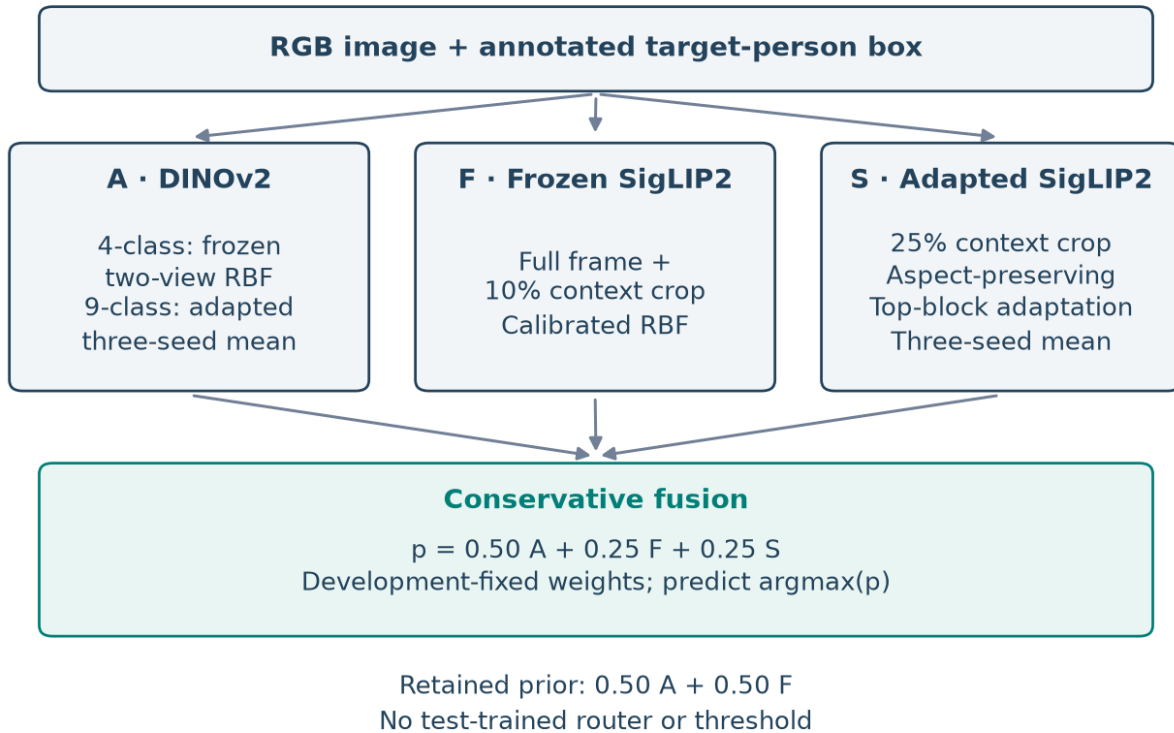
3.3 Complementary probability fusion

Let A denote the DINOv2 anchor, F frozen SigLIP2 and S the mean of the three adapted-SigLIP2 probability vectors. A is a frozen two-view classifier for four classes and a three-seed adapted model for nine classes.

System	Probability vector	Role
Prior fusion	$0.50 A + 0.50 F$	Retained incumbent
Conservative fusion	$0.50 A + 0.25 F + 0.25 S$	Development nominee
Replacement fusion	$0.50 A + 0.50 S$	Predeclared diagnostic control

Prediction is argmax of the probability vector. Weights are development-fixed; no router, threshold or annotation-only support category is learned from test labels.

Person-centric views and fixed fusion



Training labels and source groups are not inference inputs.

Figure 1. Person-centric branches and development-fixed probability fusion.

4. Locked evaluation

The final phase completed 23 refits (15 neural and eight frozen heads), followed by fixed prediction and comparison panels. Ten systems per task and 18 paired comparisons were specified before final test extraction. The selection-lock hash is recorded in the [public manifest](#).

Primary reporting uses macro-F1, supplemented by accuracy, class-wise metrics, NLL, summed multiclass Brier, calibration and selective-risk diagnostics. Uncertainty uses 5,000 class-stratified row bootstrap and 5,000 source-group bootstrap draws. Paired randomization uses 10,000 two-sided source-group swaps with the plus-one p-value correction. Holm adjustment covers all 18 comparisons. Intervals are not simultaneous or adjusted for adaptive historical selection.

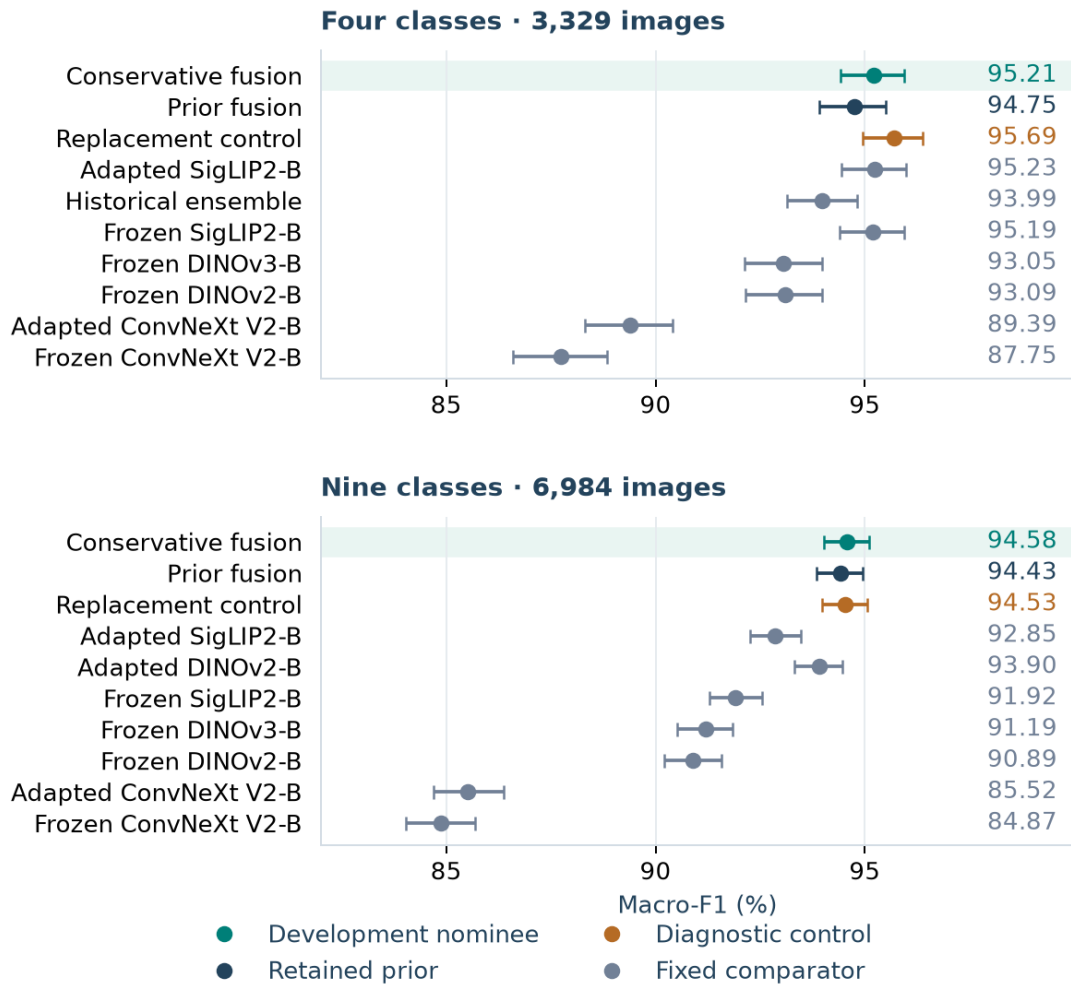
Promotion requires positive F1 and net corrections, a positive lower source-group delta interval, Holm p below 0.05, no class-F1 drop above one point, NLL increase at most 0.02, Brier increase at most 0.01, and positive F1/net corrections for every matching seed pair. The comparison family and gates remain unchanged after inspection. Engineering recoveries concerned execution and metadata, not model selection; the [dated record](#) links their audit receipts.

5. Results

The nominee makes 141 errors on four classes (95.764% accuracy) and 369 on nine (94.716% accuracy). Source-group 95% F1 intervals are 94.425–95.945% and 94.031–95.110%. The replacement control's higher four-class point estimate does not make it a post-test-selected winner. Figure 2 shows every fixed candidate; the [results tables](#) provide detailed scores and class-wise metrics.

POLAR | locked final evaluation

All predeclared systems; separate four- and nine-class tasks.



Bars: marginal 95% source-group bootstrap intervals.
Promotion uses paired tests and every locked gate.

Figure 2. All fixed comparison candidates and marginal source-group intervals.

5.1 Paired comparisons and probability quality

Nominee comparison	F1 gain (pp)	Paired 95% interval (pp)	Rescue / harm	Holm p
Four vs historical	+1.223	+0.573 to +1.940	75 / 35	0.00420
Four vs immediate prior	+0.465	+0.064 to +0.890	27 / 13	0.19858
Nine vs adapted DINOv2	+0.679	+0.381 to +0.987	80 / 33	0.00180
Nine vs immediate prior	+0.158	−0.091 to +0.408	44 / 33	0.90831

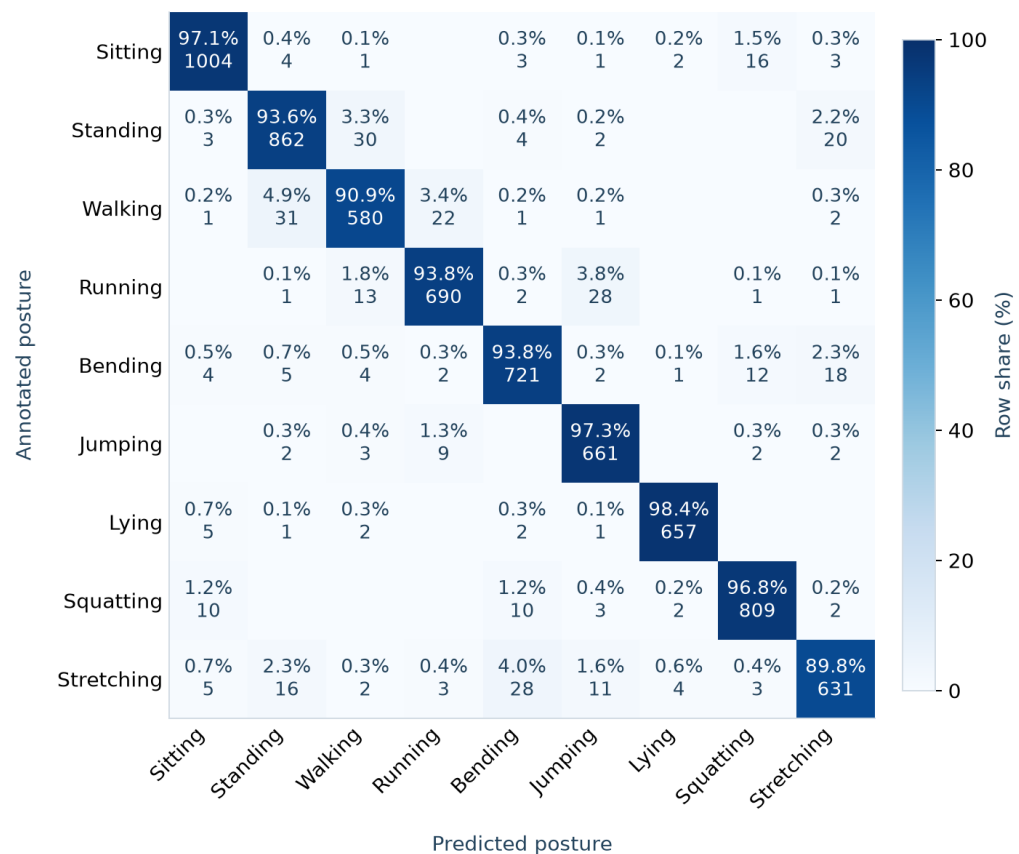
The first and third comparisons support within-protocol gains against their fixed references. Neither task passes every promotion condition against its immediate prior, despite positive seed diagnostics and passing class/probability-safety checks. The correct decision is to retain both prior models, not change the testing family or nominate the best test-scoring control.

The nominee's NLL/Brier/ECE are 0.1370/0.0664/0.0390 on four classes and 0.1684/0.0812/0.0228 on nine. Nine-class prior values are 0.2021/0.0908/0.0447. Better probability scores do not override the locked macro-F1 promotion rule. All metrics and per-class values appear in the [results guide](#).

6. Error structure and mechanism evidence

Nine-class error structure

Conservative fusion · 94.58% macro-F1 · 369 errors / 6,984 images



Cells show row-normalized percentage and count; blank cells have zero count.
Largest pairwise confusion: standing ↔ walking (61 images). No relabeling or test-driven fitting.

Figure 3. Nine-class nominee confusion. Cells show row percentage and count.

Standing/walking contributes 61 bidirectional confusions, bending/stretching 46, running/jumping 37, standing/stretching 36 and walking/running 35. Walking and stretching have the lowest nominee F1 values (91.12% and 91.32%). These describe error concentration, not proven causal mechanisms. Static ambiguity, viewpoint, background dependence and annotation semantics remain rival explanations.

Development evidence suggests three useful mechanisms. First, two-view RBF heads beat the best screened linear heads by 0.73–1.36 points on four classes and 0.75–2.29 on nine across the four families. Second, person preservation combined with the milder augmentation recipe improves the same nine-class DINOv2 seed from 90.73% to 93.23% validation F1; the two changes are not causally separated. Third, selective DINOv2/SigLIP2 fusion beats uniform all-family averaging in development. The final nine-class rescue/harm counts support complementarity directly.

The four-class result is a useful counterexample: frozen SigLIP2 already reaches 95.19%, and the nominee's 95.21% does not establish superiority over it. Model complexity is not itself evidence of improvement. These findings are documented in the [bounded development review](#).

7. Related work and limitations

The original representations are documented in the official [DINOv2 repository](#), [DINOv3 repository](#), [SigLIP 2 paper](#) and [ConvNeXt V2 repository](#). This work compares their downstream behavior rather than attributing the pretrained encoders to this project.

Related posture and adaptation papers use different taxonomies, metrics or populations. The [external comparison guide](#) distinguishes Habibi's three-class transfer evaluation, CAPA-AI's novelty adaptation and Mandita/Rokhman's multilabel posture targets. None supplies a verified directly matched nine-class baseline here. A numerical ranking across those reports would be misleading.

The given-box input, partly reused test cohort, incomplete identity/scene audit and unknown pretraining overlap constrain generalization claims. The neural three-seed means do not constitute independent test cohorts. Session-level cost records may omit earlier resumed work and feature extraction; they are not end-to-end latency or complete compute measurements. No clinical validity or demographic fairness evaluation was conducted.

8. Reproducibility and conclusion

The repository distributes training/evaluation source, audited membership, 32 probability sets and checksummed numerical reports. Public checks replay all 20 candidate metric sets, six seed pairs and 18 paired comparisons; the optional resampling pass repeats the recorded paired uncertainty and randomization tests. Full checkpoint replay requires separately obtained data and weights. [Reproduction instructions](#) and [release validation](#) separate those levels of evidence. No dataset photographs or weights are redistributed.

The study supports person-centric preprocessing, strong nonlinear frozen-feature controls and task-dependent complementary fusion. It establishes measured gains over specified internal references, preserves the prior-model promotion decisions, and provides a reproducible basis for further evaluation without presenting an unverified external benchmark rank.